ASSIGNMENT NO. 1-Study of Web and its Basics

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Problem Statement 1

1. Evolution of Web (Including Web 3.0)

The web has evolved in different phases:

- **Web 1.0 (Read Only Web):**
  Static web pages, users could only read information. No interaction.
- **Web 2.0 (Read Write Web):**
  Interactive web with social media, blogs, dynamic websites. Users can create and share content.
- **Web 3.0 (Semantic & Decentralized Web):**
  Uses AI, blockchain, and machine learning. Data is decentralized and smarter, giving personalized experiences.

2. Ports and Protocols Used by Web

Protocols:

- **HTTP:** Transfers web data
- **HTTPS:** Secure version of HTTP
- **FTP:** File transfer
- **SMTP:** Sending emails

Ports:

- HTTP → Port 80
- HTTPS → Port 443
- FTP → Port 21

Ports help identify services running on a server.

3. Difference Between HTTP and HTTPS

HTTP	HTTPS
Not secure	Secure
No encryption	Data encrypted
Port 80	Port 443
Faster but unsafe	Slightly slower but safe

4. Web Developer Tools & Need

Web developer tools are browser tools used to inspect, debug, and test websites.

Uses:

- Inspect HTML & CSS
- Debug JavaScript
- Check responsiveness
- Improve performance

Examples: Chrome DevTools, Firefox DevTools

5. Client-Server & MVC Architecture

Client-Server Architecture

- Client sends request
- Server processes and sends response

Used in simple web applications.

MVC Architecture

- **Model:** Data & logic
- **View:** UI
- **Controller:** Handles user input

When to use:

- Client-server → small apps
- MVC → large, scalable apps

6. What is HTML and HTML5?

HTML is a markup language used to create web pages.
HTML5 is the latest version with multimedia support, APIs, and better semantics.

7. Current Versions of HTML & CSS

- **HTML:** HTML5
- **CSS:** CSS3

8. Frontend & Backend Tools

Frontend Tools

- HTML, CSS, JavaScript
- React, Angular
- Bootstrap, Tailwind

Backend Tools

- Node.js
- Express
- Django
- PHP
- MongoDB, MySQL

9. MERN Stack

MERN =

- MongoDB
- Express.js
- React.js
- Node.js

Why Preferred

- Full JavaScript stack
- Fast development
- Scalable
- Used in modern web apps

10. HTML5 New Features

Input Types

- email, date, range, color

APIs

- Geolocation API
- Web Storage API

Form Elements

- datalist, progress, meter

Media Elements

- <audio>, <video>

11. HTML5 Web Storage

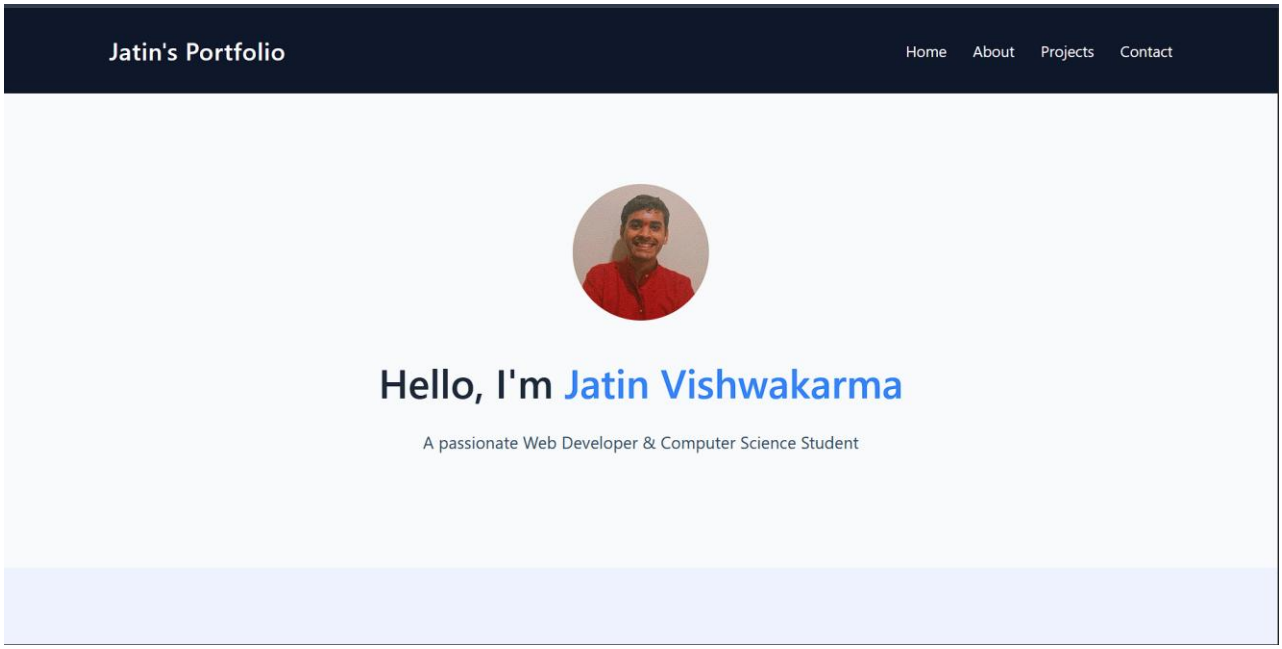
HTML5 provides:

- **Local Storage:** Permanent
- **Session Storage:** Temporary

Used to store data on client-side without cookies.

Problem Statement 2: Student Portfolio Website

OUTPUTS:



About Me

I am a second-year Computer Science and Engineering student who enjoys building simple, meaningful, and user-friendly web applications. I love learning new technologies and improving my problem-solving skills.

Education

Walchand College of Engineering, Sangli

Skills

HTML

CSS

JavaScript

Projects

Personal Portfolio

A responsive personal portfolio website showcasing my profile, skills, and projects.
Tech Stack: HTML, CSS, JavaScript
GitHub Repository

Student Management System

A basic web application to manage student records with CRUD operations.
Tech Stack: HTML, CSS, JavaScript
GitHub Repository

Weather Application

A weather app that shows real-time weather data using a public API.
Tech Stack: JavaScript, API
Live Demo

Contact Me

Your Name

Your Email

Subject

Message

Send Message

Your Name

Your Email

Subject

Message

Send Message

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